PERFORMANCE

USER GUIDE

How to train the network, perform with five knobs, and map a Launchkey.

This guide is the how-to for the device after it is installed. For install, first-load checks, and file locations, use the project **Install Guide**.

Dub Assistant sits on a MIDI track and writes Live's mixer. You teach it what **col**, **rhy**, and **dyn** mean by saving mixer snapshots, then you perform with those three knobs plus **Heart** and **Dry/Wet**.

HOW THE MAPPING WORKS

The device is a one-to-many map: three knobs in, twenty-four mixer values out. That is the mapping layer from the 2020 paper.

The twenty-four outputs are three groups of eight, one slot per named track:

Group	What it controls
Sends	Level of send A (return **Dub Delay** by default)
Slider probability	Chance a fader moves on the next 1/4 or 1/8
Slider dynamics	How far that fader drops or returns

Sends follow the knobs continuously. Faders do **not** follow a single predicted height. They move in a non-repeating way, locked to Live's transport, which is the dub-console behaviour the paper describes.

Until you click **Train**, the device uses a simple fallback map so you can hear writes immediately. After **Train**, the multilayer perceptron interpolates between the rows you saved.

PREPARE THE SET

The device finds tracks by **name**, not by mixer order. Rename, do not reorder, if you want binds to survive a rearrange.

Use these default names:

- Audio: Drums, Bass, Vox, Keys, Guitar, FX, Perc, Other
- Returns: Dub Delay (A), Dub Verb (B)
- MIDI: Dub Assistant (this device only)

Load clips however you like. If you have Live 12 Suite, run Stem Separation yourself (Live manual chapter 22), then drag the four stems onto Drums, Bass, Vox, and Other. Pal cannot trigger the split.

Put native Delay on return A and native Reverb on return B (chapter 28). The device writes send A unless you change the send index.

Click **Bind** after the names are in place. The status line reports bound N/8 tracks.

TRAIN YOUR TASTE

The empty network is not the instrument. Your dataset is.

Each **Learn this!** row stores the current **col / rhy / dyn** plus the eight send, probability, and dynamics values you see on the device. Aim for 100 or more rows. The paper's tests used that floor. Train is blocked below 8 rows.

Work through this loop:

1. Set **Dry/Wet** to 0% so you own the mixer.
2. Set **col**, **rhy**, and **dyn** to a meaning you care about. For example, high **col** = wet and dark, high **rhy** = drums and bass isolated, high **dyn** = bigger fader jumps.
3. Set the eight **send** dials and the **probability / dynamics** number boxes to the mixer behaviour you want at that point.
4. Click **Learn this!**. The row counter goes up.
5. Repeat across the extremes and the middle of the three knobs. Cover corners, not only a straight line.
6. Click **Train**. Wait for trained on N rows. Training takes a second or two for a hundred rows.
7. Click **Save** and keep the CSV next to the Live Set.

You can add rows during a performance and train again. After each **Train**, the previous network is replaced.

> TIP
>
> Load sample-dataset.csv only to confirm Train works. Then click **Clear** and record your own rows. The sample file is a geometry check, not a dub mix.

Suggested first 20 rows

Use this starter grid so the network has something to interpolate, then replace it with moves you actually like.

col	rhy	dyn	Intention
low	low	low	Dry, full band, almost still
high	low	low	Washed, full band
low	high	medium	Dry, rhythm isolated
high	high	high	Wet, rhythm isolated, busy faders
mid	mid	mid	Default dub bed

For "rhythm isolated," raise sends and dynamics on Vox / Keys / Guitar / FX, and keep Drums / Bass sends lower.

PERFORM

Performance is the five Live parameters on the left and right of the device.

1. Click **Bind**. Confirm 8/8 or at least Drums and Bass.
2. Click **Load** if you have a saved CSV, then **Train**.
3. Start the transport. Faders only tick while Live is playing.
4. Raise **Dry/Wet**. At 0% the device writes nothing. At 100% it writes the full model. In between it blends against the snapshot it took when Dry/Wet left 0%.
5. Move **col**, **rhy**, and **dyn**. Sends follow. Faders jump on 1/8 (default) or 1/4.
6. Use **Heart** to limit how much Drums and Bass may move. At 0 the rhythm section holds. The other six tracks still move.
7. Drop **Dry/Wet** to 0% the moment you want the console back.

Click **1/4** or **1/8** to change the fader grid. Eighths feel more like a live dub pass. Quarters are calmer.

> WARNING
>
> Leave **Dry/Wet** at 0% while you set up tracks, record, or edit clips. The device writes volume and send A on bound tracks whenever Dry/Wet is up and the transport is running.

MAP THE LAUNCHKEY

The five performance controls are Live parameters, so any MIDI controller can map them (Live manual chapter 33).

On a Launchkey MK4 61:

1. Select the **Dub Assistant** device in Live.
2. Enter MIDI map mode.
3. Map five knobs to **Colour**, **Rhythm**, **Dynamics**, **Heart**, and **Dry/Wet**.
4. Exit MIDI map mode.

A custom Launchkey mode is optional later. MIDI map is enough for v1.

STATUS LINE

The top of the device reports the last engine message.

Message	Meaning
ready — bind tracks, then Learn this!	Live API is up
bound N/8 tracks	Name match result
learned row N	Dataset grew
trained on N rows	Network replaced
need at least 8 rows	Train blocked
saved N rows / loaded N rows	File I/O
faders on 1/4 or 1/8	Grid changed

If bind returns 0/8, the names do not match. Rename the audio tracks or, from the Max window, send:

names Drums Bass Vox Keys Guitar FX Perc Other

To treat different tracks as Heart (the ones Heart scales):

heartnames Drums Bass

To write send B instead of send A:

sendindex 1

SAVE THE SET

After the device is trained and mapped:

1. Save the dataset CSV next to the Set, not only inside Live.
2. Use **File → Collect All and Save** so clips and the device travel together (Live manual chapter 5).
3. If you share the device, freeze it from the Max toolbar (snowflake) and keep dub-assistant.js listed in the device's project. The production guidelines in the Max for Live manuals cover freeze and dependencies.

NEXT STEPS

Keep **Operator Card** visible while you play. Use **Template Set** when you start a new piece from stems.